

CSC11006 - INTRODUCTION TO CLOUD COMPUTING SERVICES

PRACTICE

LAB2: Pricing_calculate_and_Cost_Management in Azure

I. General information

ID: LAB12
Period: 30 minutes
Deadline:

II. Outcome:

This lab will adapt to these following course's outcomes:

- G2.1
- G2.2

III. Describe:

Explore the Pricing calculator

1. Define your requirement:

In this exercise, you use the Pricing calculator to estimate the cost of running a basic web application on Azure. Start by defining which Azure services you need.

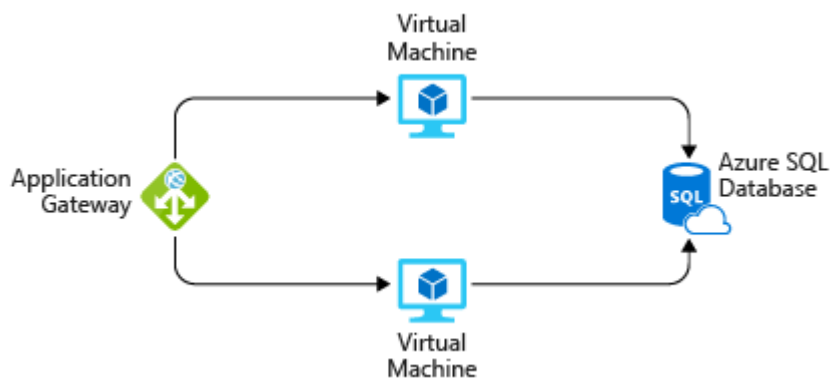
For a simple web application hosted in your datacenter, you might have a setup like this:

An ASP.NET web application running on Windows, providing information about product inventory and pricing. It is supported by two virtual machines (VMs) connected through a central load balancer. The web application connects to a SQL Server database that stores inventory and pricing data.

To migrate this setup to Azure, you could:

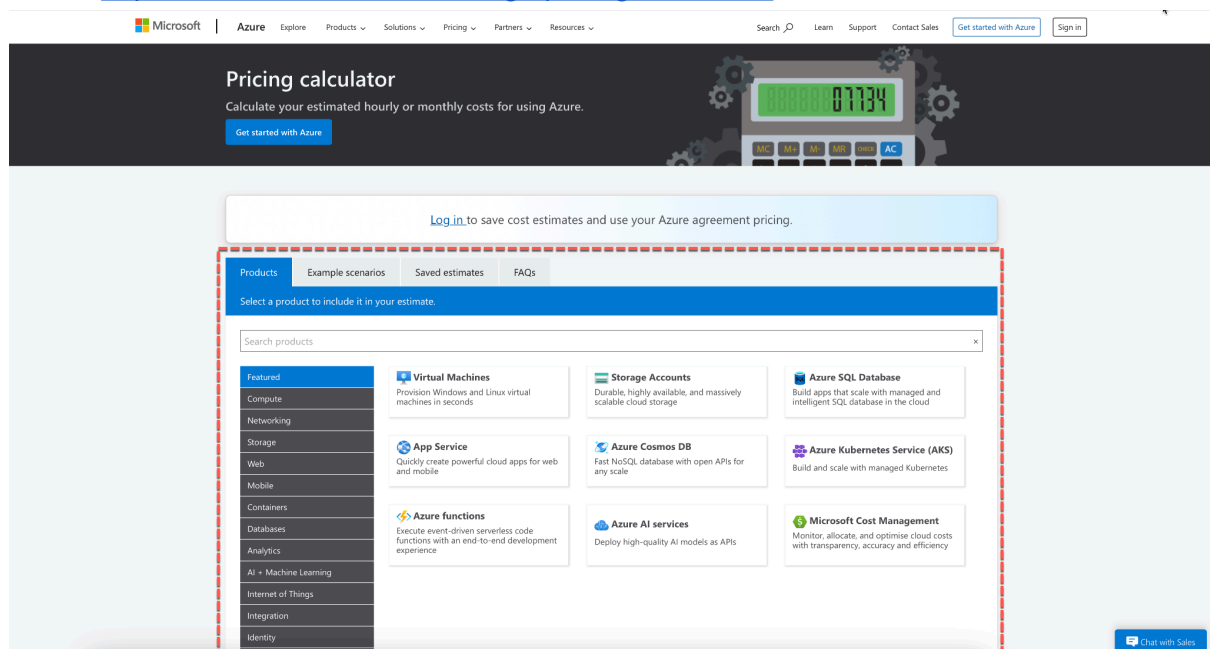
- Use Azure Virtual Machines, similar to the VMs in your datacenter.
- Implement Azure Application Gateway for load balancing.
- Move the database to Azure SQL Database to store inventory and pricing information.

This is the diagram that shows basic configuration:



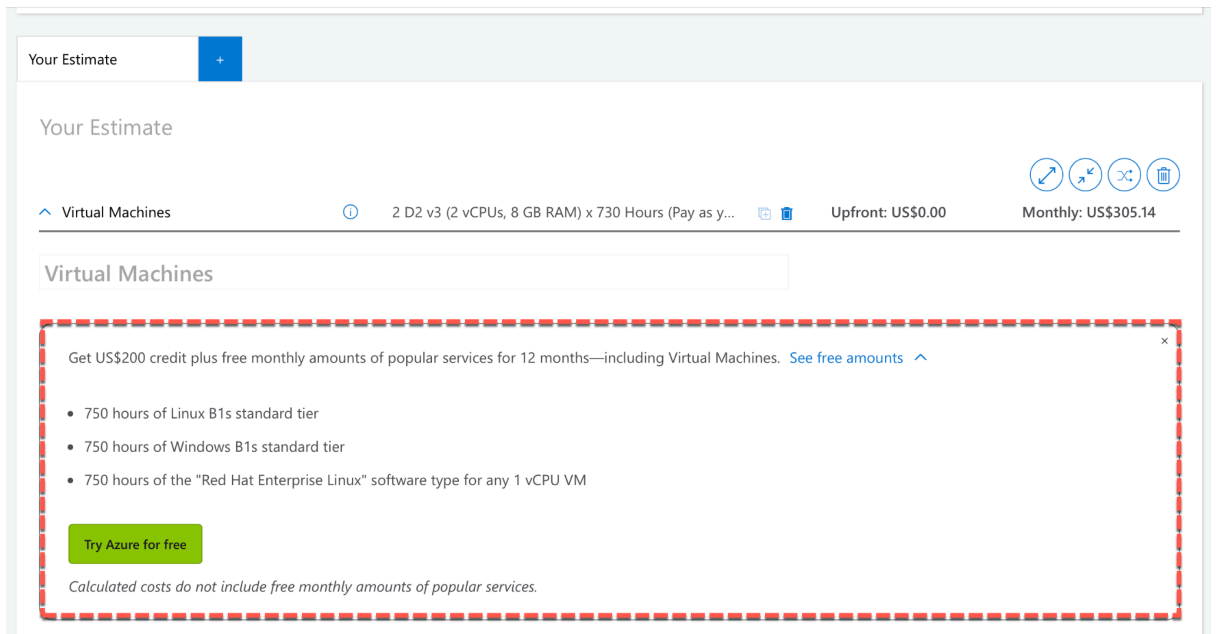
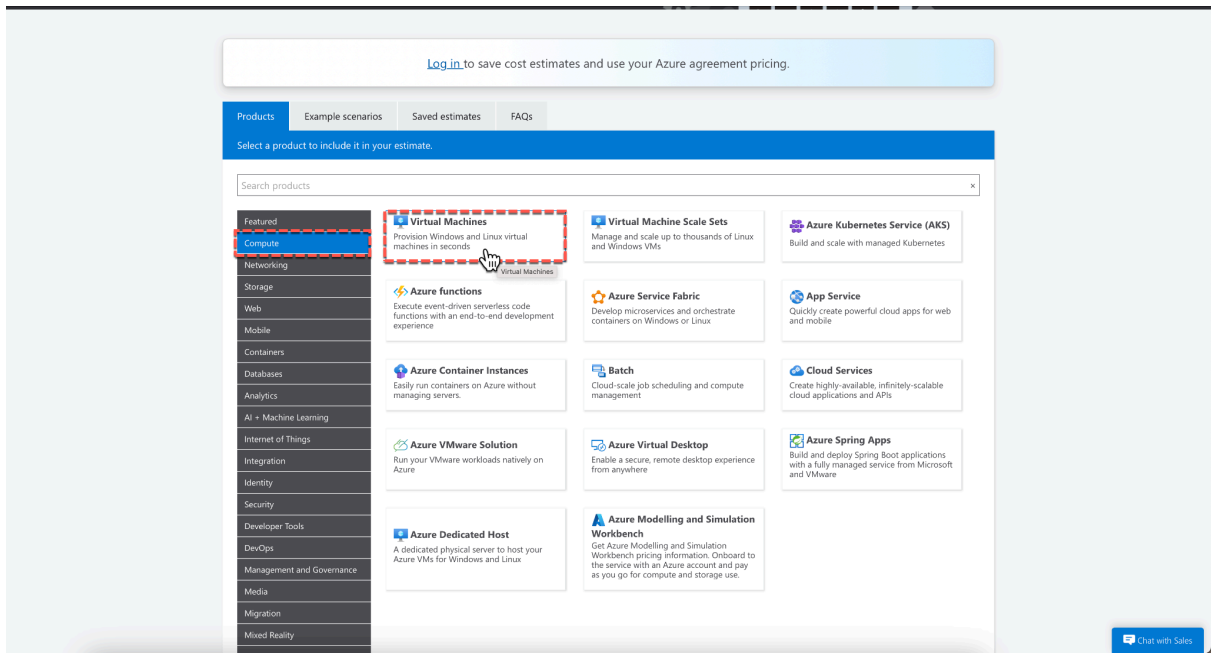
2. Access the Pricing calculator page:

<https://azure.microsoft.com/en-gb/pricing/calculator/>



3. Estimate your solution:

3.1. Virtual machine:



Virtual Machines
2 D2 v3 (2 vCPUs, 8 GB RAM) x 730 Hours (Pay as y...
Upfront: US\$0.00
Monthly: US\$305.14

Virtual Machines

Region: West US
Operating system: Windows
Type: (OS Only)
Tier: Standard

Category: All
Instance series: Dv3-series
INSTANCE: [\(Need help finding the right VM?\)](#)
D2 v3: 2 vCPUs, 8 GB RAM, 50 GB Temporary storage, US\$0.209/hour

2 x 730 Hours

Virtual machines

Savings options

Explore pricing models to help optimise your Azure costs. [Learn more](#)

Compute (D2 v3)

☒ Pay as you go

OS (Windows)

☒ Licence included

☐ Azure Hybrid Benefit

Savings plan

☐ 1 year savings plan (~31% discount)

☐ 3 year savings plan (~53% discount)

Reservations

☐ 1 year reserved (~32% discount)

☐ 3 year reserved (~57% discount)

US\$170.82
Average per month
(US\$0.00 charged upfront)

US\$134.32
Average per month
(US\$0.00 charged upfront)

= US\$305.14
Average per month
(US\$0.00 charged upfront)

3.2. Database:

[Log in](#) to save cost estimates and use your Azure agreement pricing.

Products
Example scenarios
Saved estimates
FAQs

Select a product to include it in your estimate.

Search products

Featured
Compute
Networking
Storage
Web
Mobile
Containers
Databases
Analytics
AI + Machine Learning
Internet of Things
Integration
Identity
Security
Developer Tools
DevOps
Management and Governance
Media
Migration
Mixed Reality
Hybrid + multcloud

Azure Cosmos DB
Fast NoSQL database with open APIs for any scale

Azure SQL Database
Build apps that scale with managed and intelligent SQL database in the cloud

Azure Database for MySQL
Fully managed, scalable MySQL database

Azure Database for MariaDB
Managed MariaDB database service for app developers

Azure Database for PostgreSQL
Innovate faster with fully managed, intelligent, and scalable PostgreSQL

Azure Synapse Analytics
Limitless analytics service with unmatched time to insight

Azure Database Migration Service (classic)
Accelerate your data migration to Azure

Azure Cache for Redis
Accelerate apps with high-throughput, low-latency data caching

Azure Health Data Services
Unify and manage health data and protected health information (PHI) in the cloud

Azure SQL Managed Instance
Migrate, modernise, and innovate with the modern SQL family of cloud database services

Azure Managed Instance for Apache Cassandra
Cloud Cassandra with flexibility, control and scale

Azure Managed Confidential Consortium Framework
Build enterprise-grade, consortium-style stateful services and decentralised-trust applications with a common source of truth.

Azure Managed Redis
Fully-managed in-memory datastore delivering the latest Redis innovations, designed for hyperscale cloud environments

Chat with Sales

Azure SQL Database

Get US\$200 credit plus free monthly amounts of popular services for 12 months—including Azure SQL Database. [See free amounts](#)

Region:
West US

Type:
Single Database

Purchase model:
vCore

Service Tier:
General Purpose

Compute Tier:
Provisioned

Hardware type:
Standard-series (Gen 5)

Instance:
8 vCore

Disaster Recovery:
Primary or Geo replica

Compute

Redundancy: Locally Redundant

1 Databases × 730 Hours

Savings options

Save up to 73% on pay as you go prices with 1 year or 3 year reserved options.

Compute

☒ Pay as you go

Reservations

☐ 1 year reserved

☐ 3 year reserved

SQL Licence

☒ Pay as you go

☐ Azure Hybrid Benefit

☐ Failover rights, standby replica

US\$977.84	US\$583.80	US\$1,561.65
Average per month	Average per month	Average per month
(US\$0.00 charged upfront)	(US\$0.00 charged upfront)	(US\$0.00 charged upfront)

3.3. Application Gateway

Products
Example scenarios
Saved estimates
FAQs

Select a product to include it in your estimate.

Search products

Featured

Compute

Networking

Storage

Web

Mobile

Containers

Databases

Analytics

AI + Machine Learning

Internet of Things

Integration

Identity

Security

Developer Tools

DevOps

Management and Governance

Media

Migration

Mixed Reality

Hybrid + multcloud

Microsoft Defender for Cloud

Protect your multi-cloud and hybrid environments

Microsoft Sentinel

Put cloud-native SIEM and intelligent security analytics to work to help protect your enterprise

Azure Dedicated HSM

Manage hardware security modules that you use in the cloud

Defender External Attack Surface Management

Safeguard the digital experience by discovering all internet-exposed resources with Microsoft Defender External Attack Surface Management (EASM)

Trusted Signing

Secure your applications through a fully managed end-to-end signing service for code, documents, applications, and more

Azure DDoS Protection

Protect your Azure resources from distributed denial-of-service (DDoS) attacks

VPN Gateway

Establish secure, cross-premises connectivity

Application Gateway

Build secure, scalable, highly available web front ends in Azure

Azure Managed Confidential Consortium Framework

Build enterprise-grade, consortium-style stateful services and decentralised-trust applications with a common source of truth.

Microsoft Entra ID (formerly Azure AD)

Synchronise on-premises directories and enable single sign-on

Key Vault

Safeguard and maintain control of keys and other secrets

Azure confidential ledger

Tamperproof, unstructured data store hosted in trusted execution environments (TEEs) and backed by cryptographically verifiable evidence

Microsoft Copilot for Security

Protect at the speed and scale of AI with a generative AI-powered assistant

Chat with Sales

Application Gateway

Region: West US Tier: Web Application Firewall Size: Medium

① No charge for the first 10 TB of data processed for a Medium instance.

Gateway hours

2 Instances × 730 Hours = **US\$206.04**

Data processed

1 TB = **US\$0.00**

Outbound Data Transfer

5 GB = **US\$0.00**

Upfront cost	US\$0.00
Monthly cost	US\$206.04

4. Review the total estimate cost:

Support

SUPPORT: Basic (Included) ① **US\$0.00**

Select your program/offer

LICENSING PROGRAMME: Microsoft Customer Agreement (MCA) ① [Log in](#) to see your Azure agreement pricing.

☒ Show Dev/Test Pricing ①

Estimated upfront cost	US\$0.00
Estimated monthly cost	US\$2,078.56

[Export](#)
[Save](#)
[Share](#)
[Log in](#) to save and share cost estimates.

CURRENCY: United States - Dollar (\$) USD

At this stage, you have several options:

- **Export** to save your estimate as an Excel file.
- **Save** or **Save As** to store your estimate in the Saved Estimates tab for future reference.
- **Share** to generate a URL, allowing you to share the estimate with your team.

Cost Management

You can access the Cost Management tools from various locations in the Azure portal

P9-Real Hands-On Labs ☆ ...
Subscription

Search Cancel subscription Rename → Change directory Feedback

Overview

- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Security
- Events
- Cost Management
 - Cost analysis
 - Cost alerts
 - Budgets
 - Advisor recommendations
- Billing
- Settings
- Help

Essentials

Subscription ID : 28e1e42a-4438-4c30-9a5f-7d7b488fd883
 Directory : Pluralsight Cloud (realhandsonlabs.com)
 My role : Resource access
 Offer : Enterprise Agreement
 Offer ID : Unauthorized
 Parent management group : ProdSubscriptions

Subscription name : [P9-Real Hands-On Labs](#)
 Current billing period : Unauthorized
 Currency : Unauthorized
 Status : Active
 Secure Score : [Not available](#)

Top products by number of resources

View resources

Azure Defender coverage

Azure Defender is not enabled for this subscription
[Upgrade coverage](#)

Cost Analysis

- **Navigate to Cost Analysis:** In the **Cost Management** section, click on **Cost analysis**.
- **View Your Costs:** Here you can see a graphical representation of your spending over time. You can filter by:
 - **Subscription:** Choose a specific subscription.
 - **Resource Group:** View costs at the resource group level.
 - **Service or Resource:** Break down costs by individual Azure services or resources.
- **Customize the View:** Use filters, date ranges, and groupings to dive deep into specific cost patterns.

Microsoft Azure

Home > Resource groups > 929-72087f89-cost-planning-and-management-in-azure | Cost analysis

Resource groups

929-72087f89-cost-planning-and-management-in-azure | Cost analysis

Save Save as Delete view Share Subscribe Refresh Download Cost by resource Configure resource group Feedback Try preview Help

Scope: 929-72087f89-cost-planning-and-man... AccumulatedCosts Jan 2023

ACTUAL COST FORECAST UNAVAILABLE BUDGET

Group by: None Granularity: Accumulated Area

No cost reported during this period — [Go to last month](#)

Service name Location Resource

Nothing to display

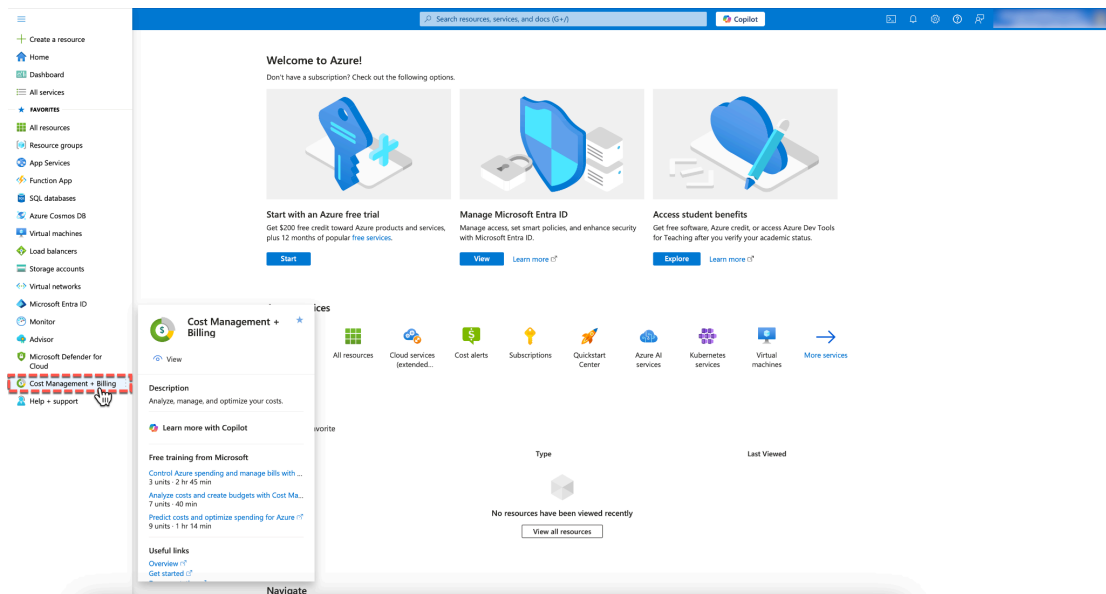
Page 1 of 1

Build budgets and alerts

1. Create a budget:

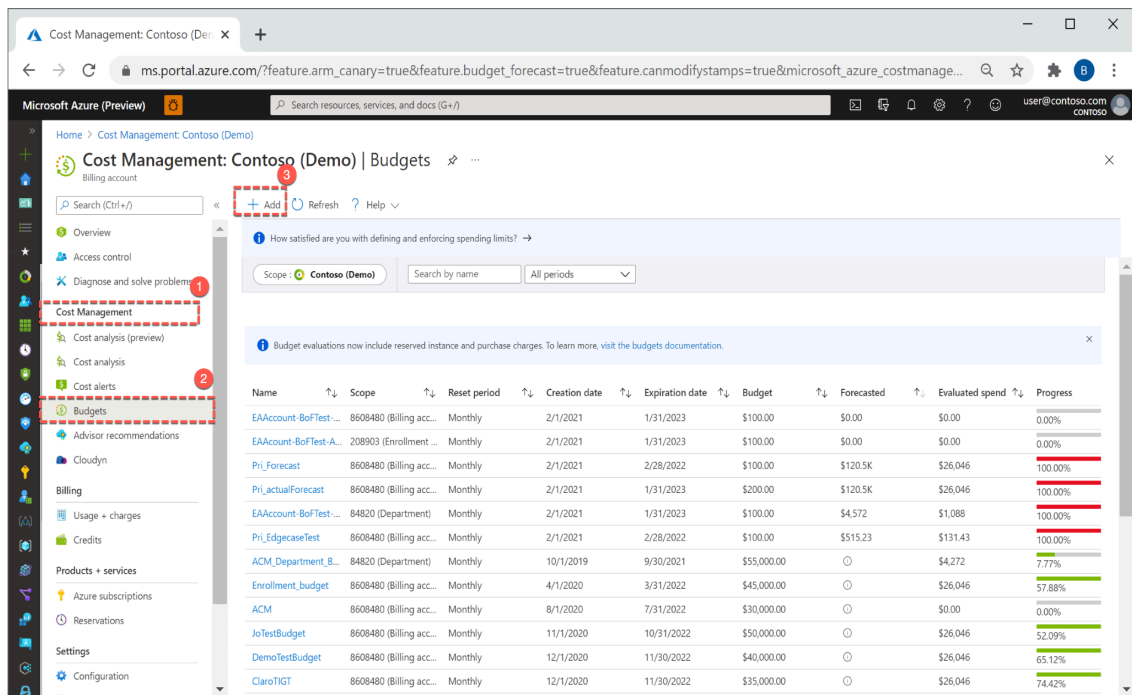
Open your browser and go to the [Azure Portal](#). Sign in using your Microsoft account or Azure Active Directory credentials.

In the left-hand menu, search for **Cost Management + Billing**. Click on **Cost Management + Billing** to open the service.



In the **Cost Management** section, select **Budgets** from the menu on the left side.

If this is your first time setting up a budget, you'll see a prompt to create one. Otherwise, click on **+ Add** to create a new budget.



2. Configure your budget:

- **Select the scope** of your budget (Subscription, Resource Group, or Management Group).
- **Set a budget name** (e.g., "Monthly Cloud Budget").
- **Choose a budget amount** (e.g., \$500/month).
- **Set the time period** for the budget (e.g., Monthly, Quarterly, or Annually).

Budget scoping

The budget you create will be assigned to the selected scope. Use additional filters like resource groups to have your budget monitor with more granularity as needed.

Scope: Contoso (Demo) [Change scope](#)

Filters: [Add filter](#)

Budget Details

Give your budget a unique name. Select the time window it analyzes during each evaluation period, its expiration date and the amount.

* Name: ✓

* Reset period: Monthly

* Creation date: 2021 March 1

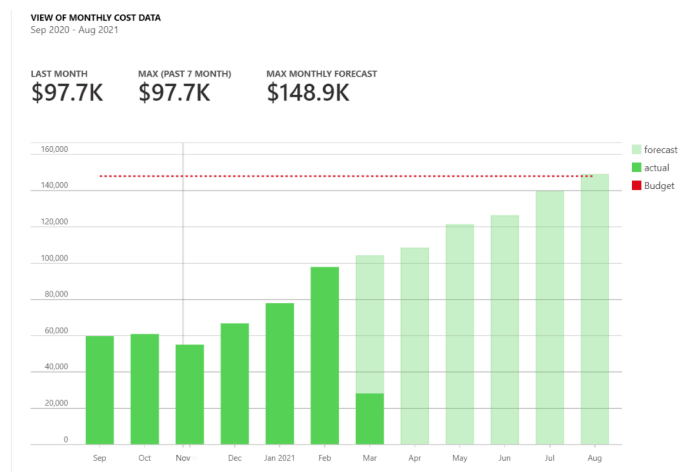
* Expiration date: 2023 February 28

Budget Amount

Give your budget amount threshold

Amount (\$) *: ✓

[Suggested budget: \\$148.9K based on forecast.](#)



[Previous](#) [Next >](#)

- After you configure the budget amount, select **Next** to configure budget alerts for actual cost and forecasted budget alerts. (Optional)

- **Set up notifications** to alert you when your spending approaches or exceeds the set budget. Alert limits support a range of 0.01% to 1000% of the budget threshold.

For example, you can set up alerts at 80%, 90%, and 100% of your budget to get email notifications.

The screenshot displays the 'Cost Management: Trey Research - Cost alerts' page in the Microsoft Azure portal. The left-hand navigation pane shows the 'Cost alerts' option highlighted with a red dashed box. The main content area shows a list of active alerts, also highlighted with a red dashed box. The table below lists the alerts:

TYPE	NAME	DATE	STATUS	SCOPE
Budget - cost exceeded th...	Actual greater than 90% (Presupuestolnno...	Thursday, February 7, 2019	Active	Trey Research Finance (Subsc...
Budget - cost exceeded th...	Actual greater than 30% (test, January 2019)	Wednesday, February 6, 2019	Active	Trey Research Corporate (Sub...
Budget - cost exceeded th...	Actual greater than 90% (TeamBudget, Januar...	Wednesday, February 6, 2019	Active	Trey Research Finance (Subsc...
Budget - cost exceeded th...	Actual greater than 90% (Monthly, January 20...	Wednesday, February 6, 2019	Active	Trey Research Finance (Subsc...
Budget - cost exceeded th...	Actual greater than 50% (Deltabudget, Januar...	Tuesday, February 5, 2019	Active	Trey Research HR (Subscripti...
Budget - cost exceeded th...	Actual greater than 90% (MonthlyBudget1, Ja...	Tuesday, February 5, 2019	Active	Trey Research Finance (Subsc...
Budget - cost exceeded th...	Actual greater than 50% (MonthlyBudget2, Ja...	Tuesday, February 5, 2019	Active	Trey Research Finance (Subsc...

Below the table, the 'Alert Details' panel is shown for the first alert. It includes a description, current cost, last update, and other relevant information.

3. Review and Create

- Review all the settings you configured.
- Click **Create** to set up the budget.

4. Clean up resources

If you created a budget and you no longer need it, view its details and delete it.